

MATTHEW A. SHAW

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EDUCATION

UNIVERSITY OF UTAH

Salt Lake City, UT

Bachelor of Science in Computer Science

Expected May 2027

- **Relevant Coursework:** Intro to Algorithms and Data Structures, Software Practice I & II, Computer Organization, Models of Computation, Database Systems, Algorithms, Mobile App Programming

ACADEMIC PROJECTS

SOFTWARE PRACTICE (CS 3500 & CS 3505)

- Developed a multiplayer Snake application in C# / Blazor using TCP client-server communication, asynchronous network updates, and a graphical MVC-based interface.
- Built C++ desktop applications with Qt, particularly a team-developed educational application.
- Collaborated through pair programming and larger development teams using Git/GitHub, branching, pull requests, and iterative Agile development.

DATABASE SYSTEMS (CS 5530)

- Designed a relational schema for LMS supporting departments, courses, classes, users, enrollment, assignments, submissions, and grading.
- Translated an ER model into normalized SQL tables with primary/foreign keys, constraints, and relationships.
- Integrated the database with a C# web application using Entity Framework and LINQ.

MOBILE APP PROGRAMMING (CS 4530)

- Developed a multi-screen Android drawing application in Kotlin/Jetpack Compose with three collaborators, supporting customizable drawing tools, image sharing/import, and responsive phone/tablet layouts.
- Implemented MVVM and repository architecture with Room persistence and Jetpack Navigation for application state, local data storage, and screen transitions.
- Built automated unit tests and collaborated through Git-based version control and team code integration.

INDEPENDENT PROJECTS

GAME STAT OCR OVERLAY (C#, WPF)

- Developed a Windows desktop overlay in C# / WPF that extracts on-screen game statistics using OCR and displays parsed results in real time.
- Built fuzzy parsing and validation logic to convert imperfect OCR output into structured data.
- Optimized the capture, parsing, and display pipeline to reduce update latency and memory usage.

TECHNICAL STACK PROFICIENCIES

- **Languages:** C#, C++, Java, Kotlin, Python, SQL, HTML/CSS.
- **Technologies/Tools:** Git, GitHub, Jetpack Compose, WPF, Entity Framework, Android Studio

OTHER WORK EXPERIENCE

ARCTIC CIRCLE

Highland, UT

Shift Supervisor

June 2022-Present

- Supervised and trained new team members in shift operations and customer service practices.
- Managed closing operations, balancing speed and accuracy to consistently exceed nightly performance targets.
- Collaborated with management to create improved checklists that reduced training time and increased accuracy.